

Submission & Reviewer Matching

Weekly update from Qiang

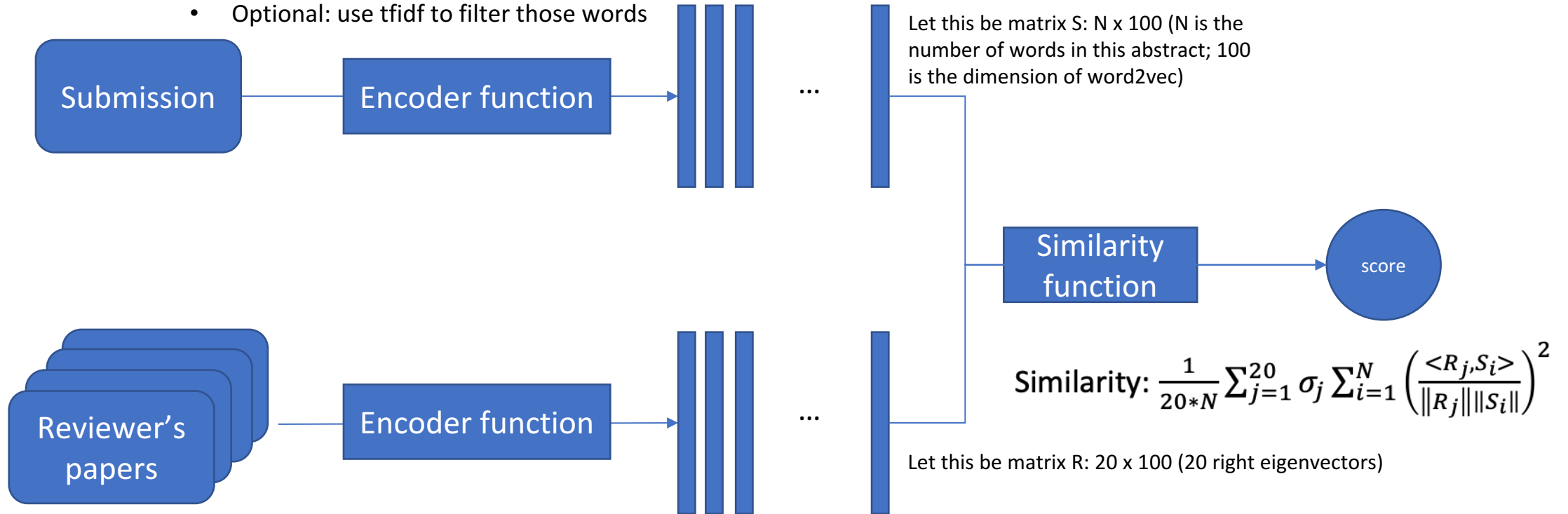
07/30/2019

Update

1. Initial try of TFIDF
2. Evaluation metric
3. Word embeddings
4. Topic weights
5. Re-organized the code
6. Summary of previous results
7. Initial try of ELMo

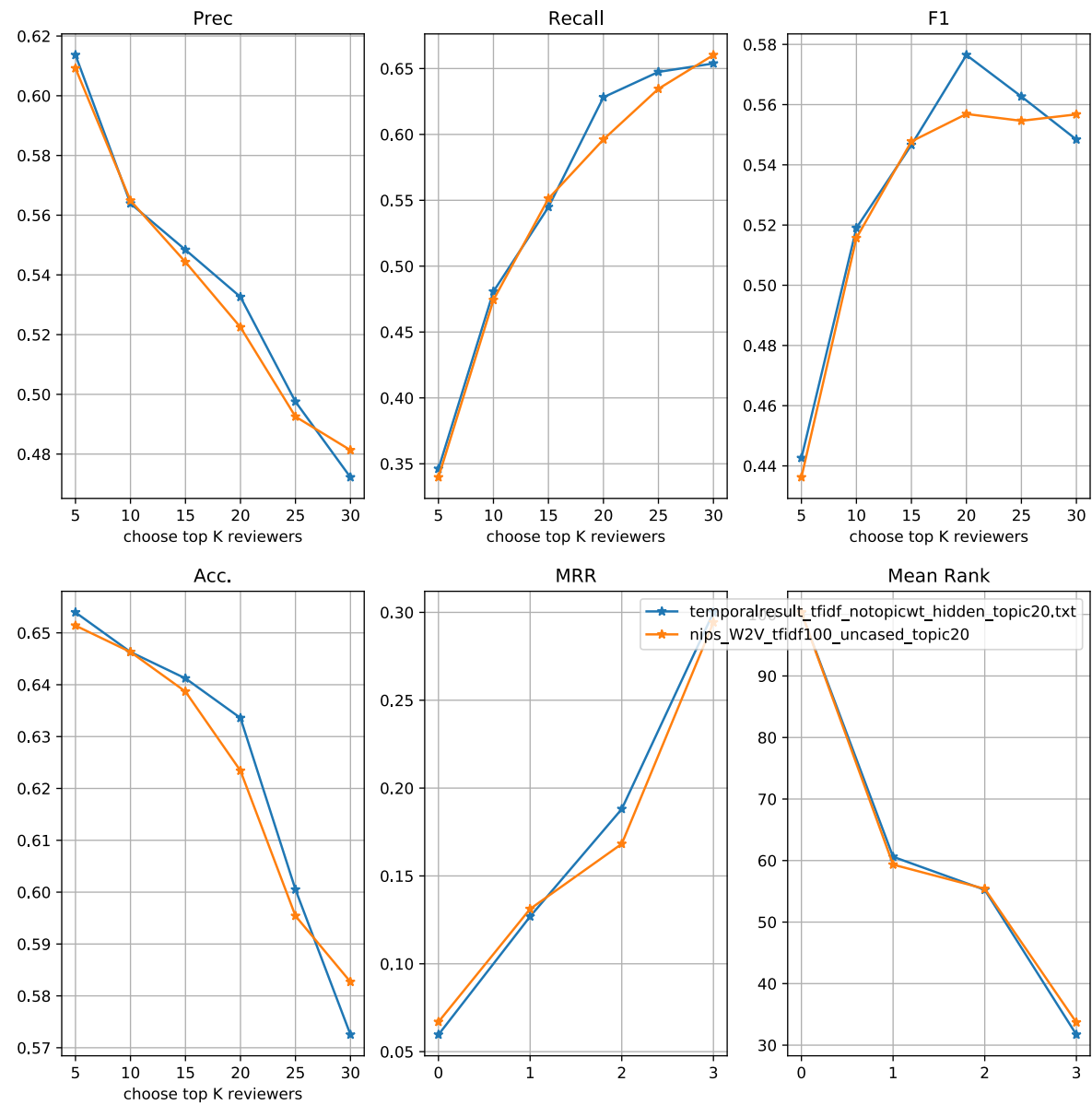
What we have now

- Pre-train word2vec on all the papers in our corpus
- Stack all the word vectors from the abstract of the submission
- Optional: use tfidf to filter those words

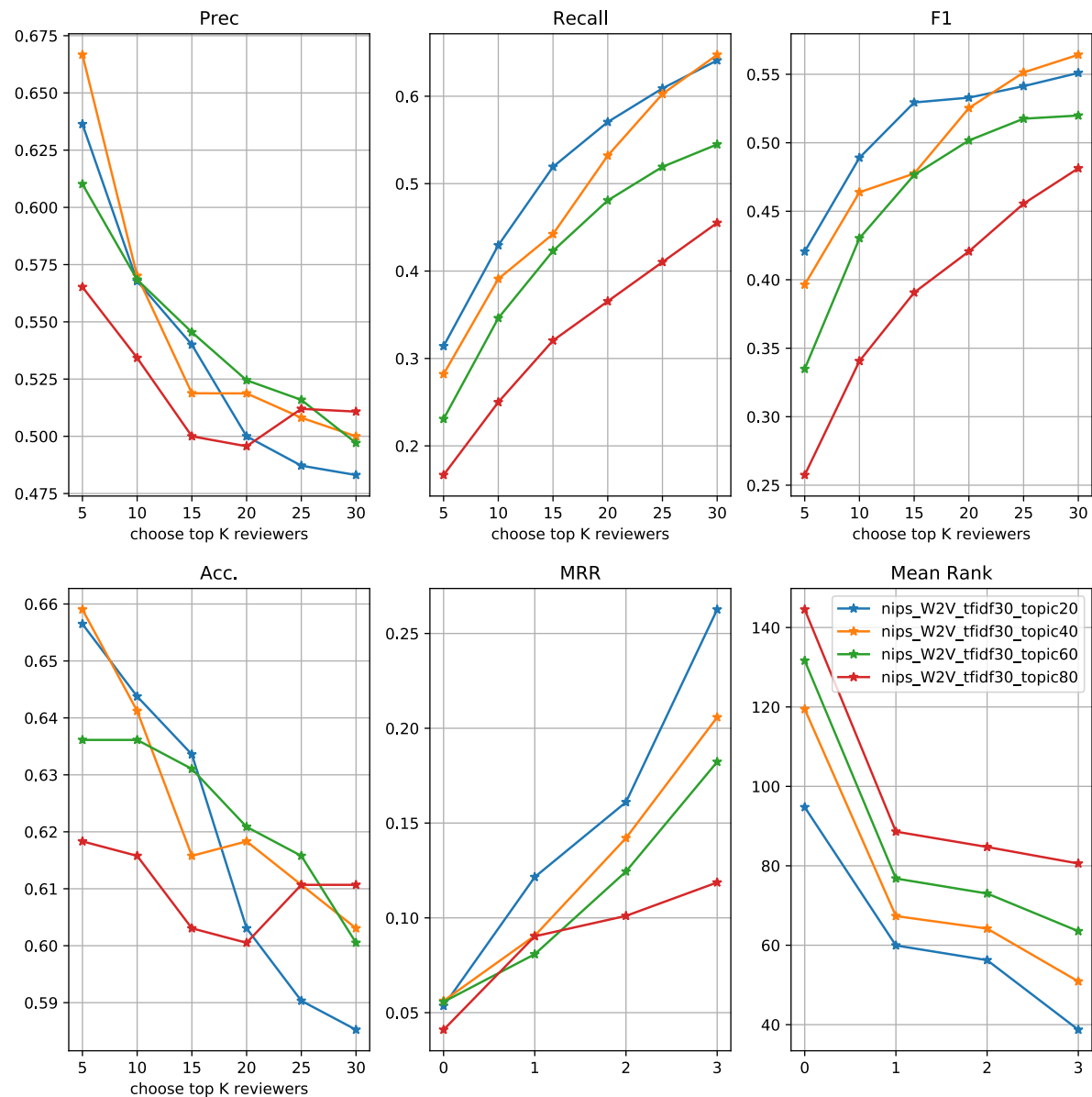


- Pre-define a weight vector for papers in specific years (putting more weight on more recent papers)
- Stack all the word vectors from the abstract of the papers of one particular review
- Perform SVD and obtain the first 20 right eigenvectors
- Optional: use tfidf to filter those words

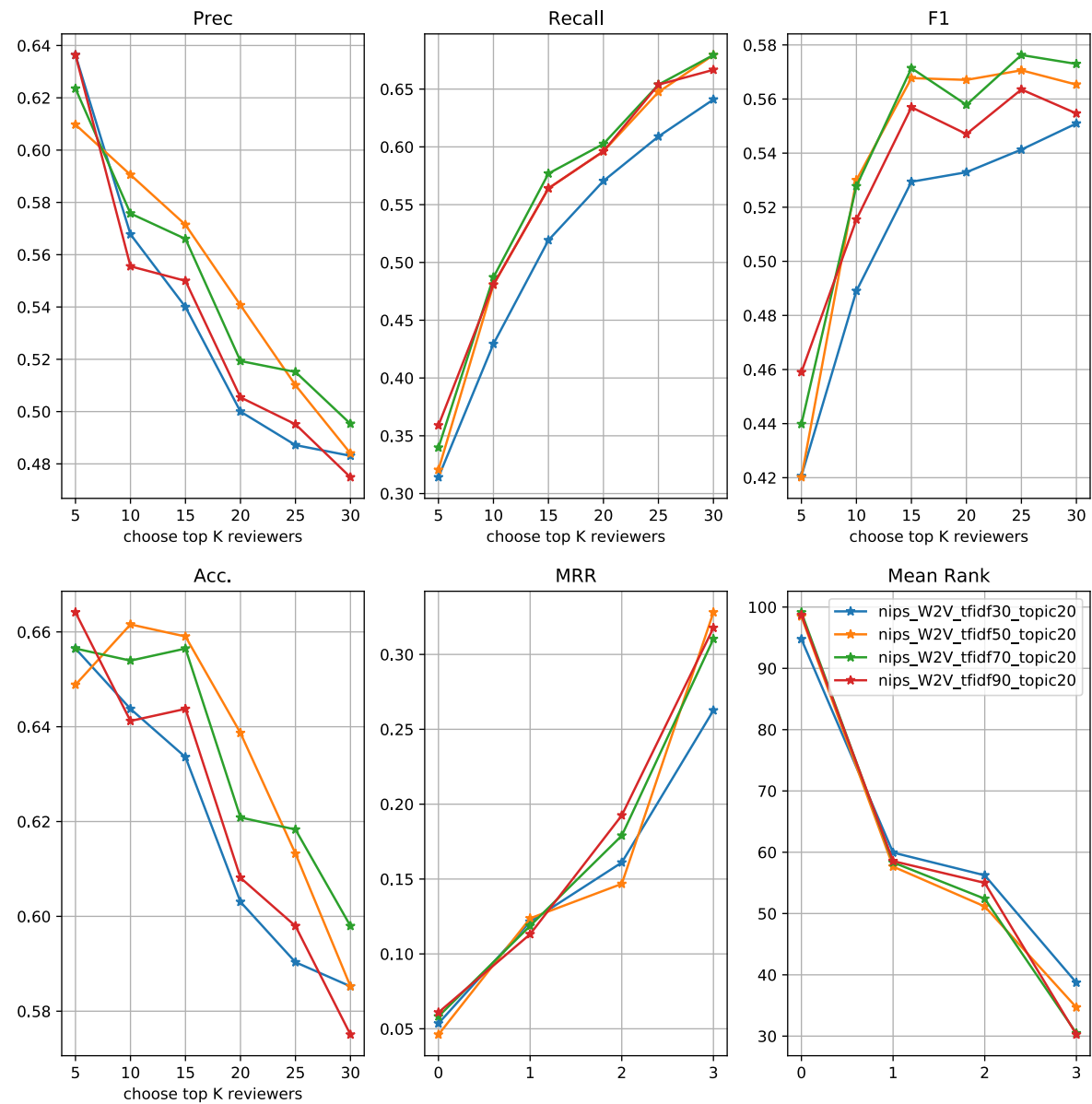
Before and after code reorganization: They're roughly the same. I currently don't know the source of difference.



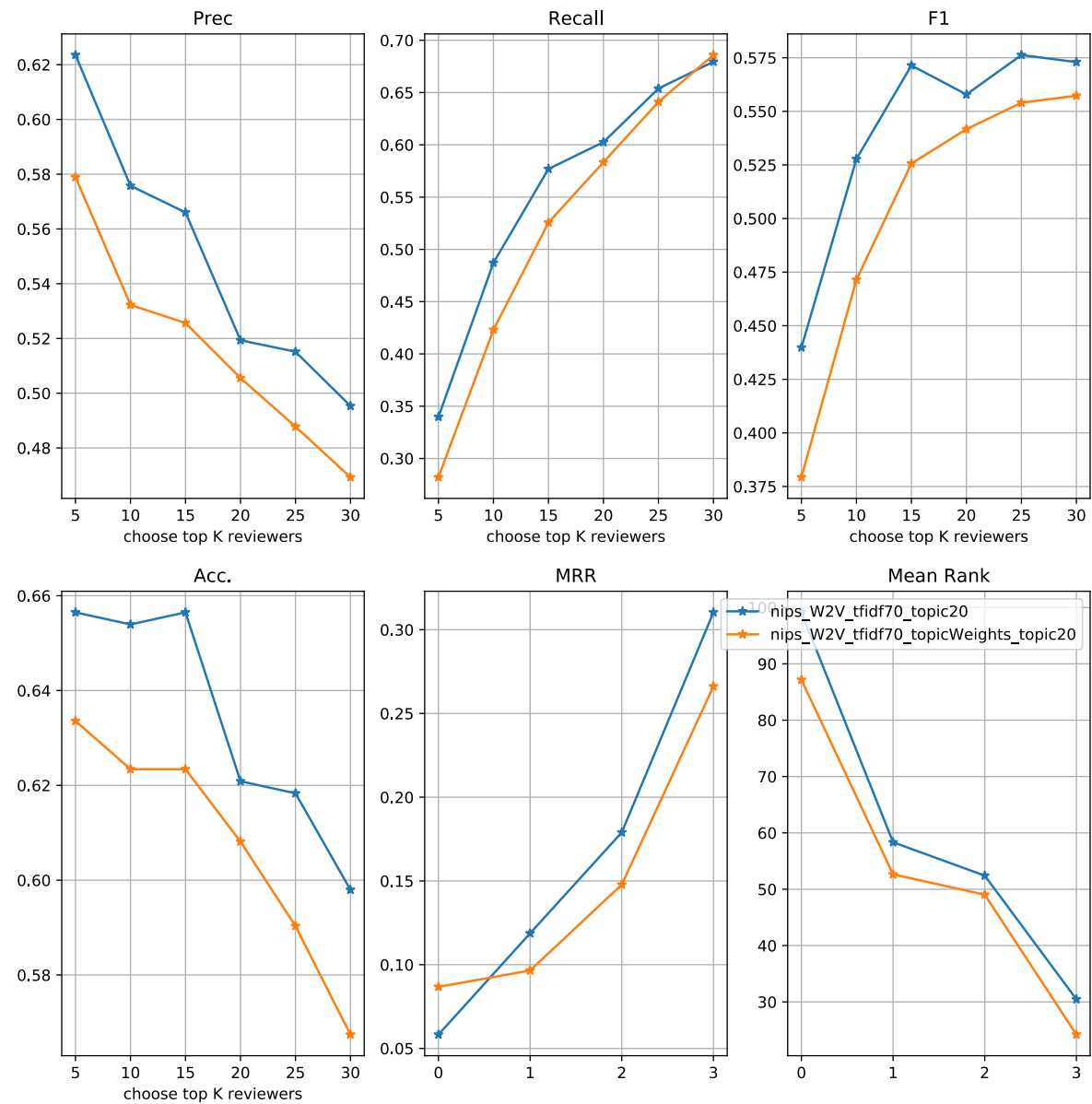
Using different topic numbers (i.e., rank of SVD): topic 20 seems to be the best option



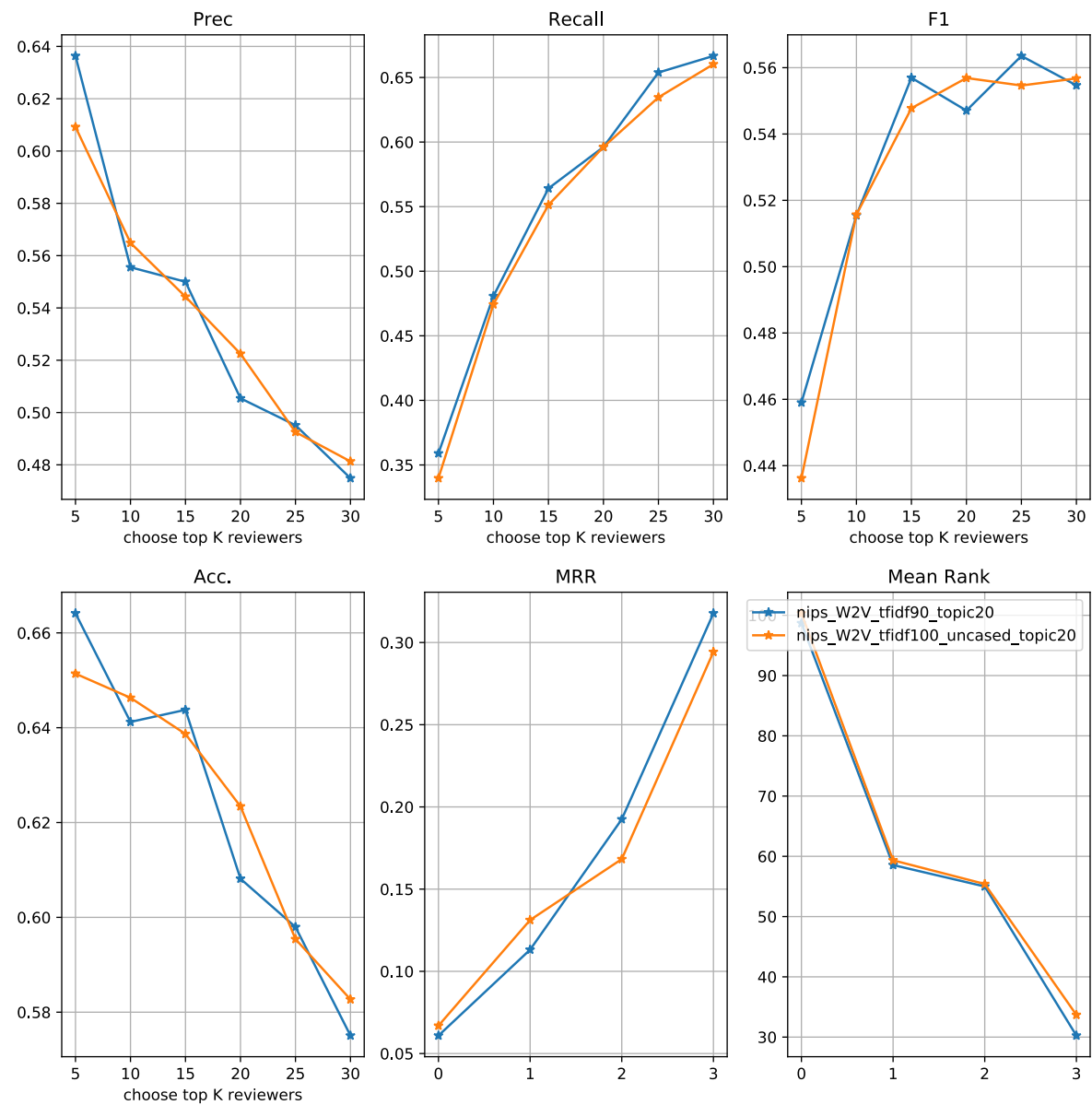
Using different TFIDF word limits: it seems that 50 or 70 is the best option for word limit



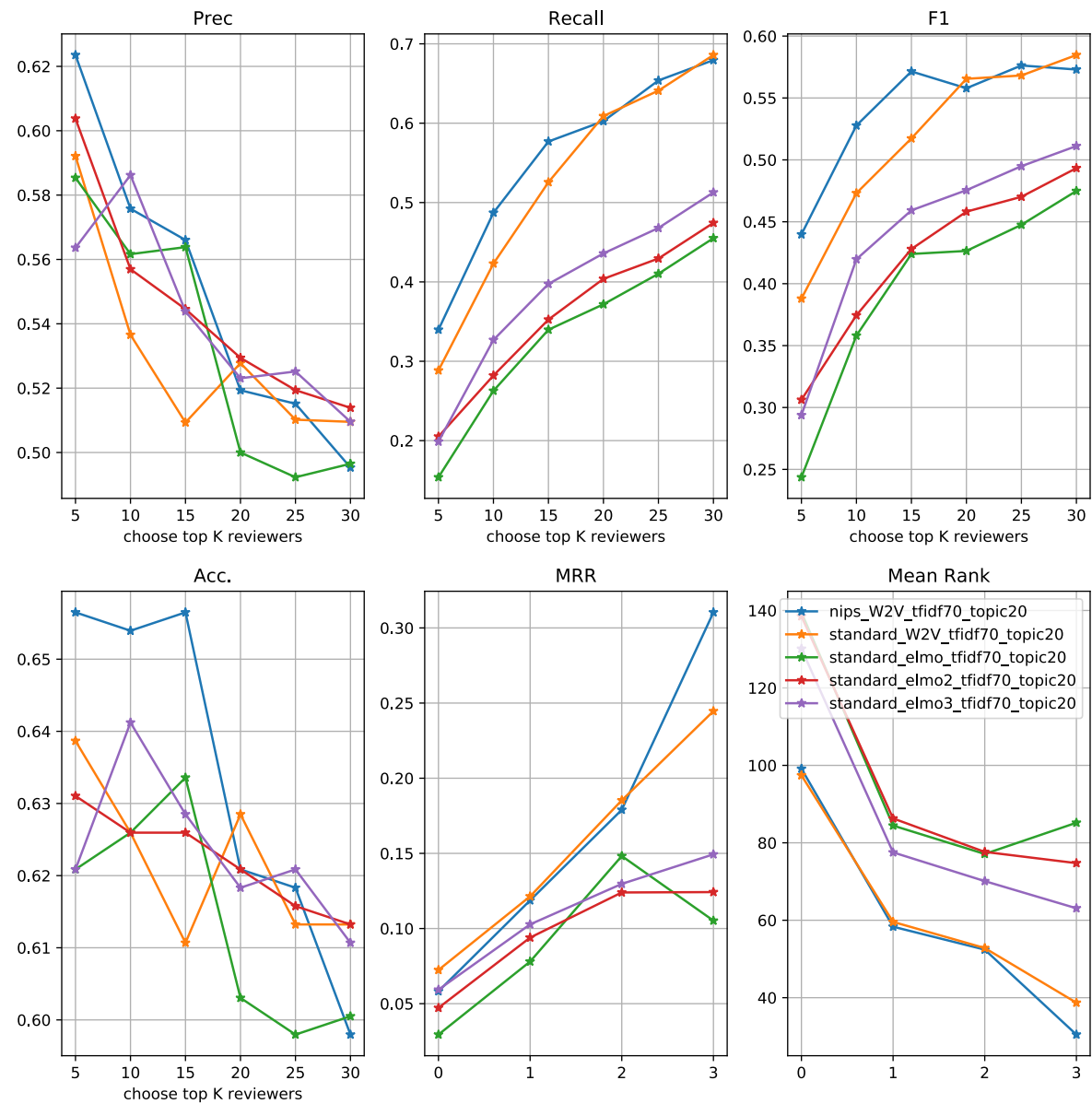
Whether or not to use topic weights: it's consistently a better option not to use topic weights



Whether or not to be case sensitive: roughly the same



Different embeddings: nips word2vec is generally the best; ELMo consistently lower.



Potential explanation of ELMo

- “a key issue in supervised protein classification is the representation of input sequences of amino acids recent work using string kernels for protein data has achieved state-of-the-art classification performance however, such representations are based only on labeled data examples with known 3d structures, organized into structural classes while in practice, unlabeled data is far more plentiful in this work, we develop simple and scalable cluster kernel techniques for incorporating unlabeled data into the representation of protein sequences we show that our methods greatly improve the classification performance of string kernels and outperform standard approaches for using unlabeled data, such as adding close homologs of the positive examples to the training data we achieve equal or superior performance to previously presented cluster kernel methods while achieving far greater computational efficiency”